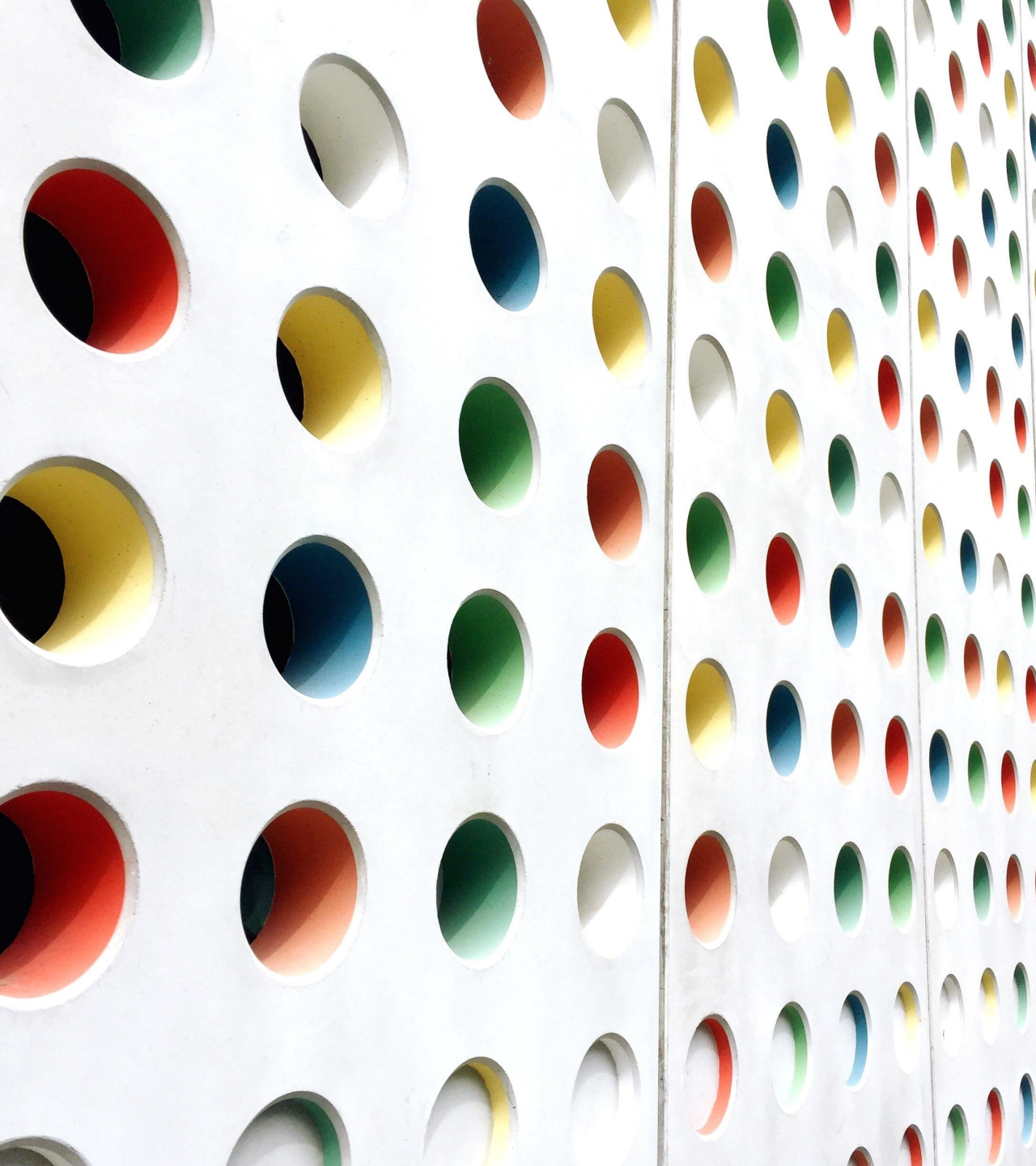
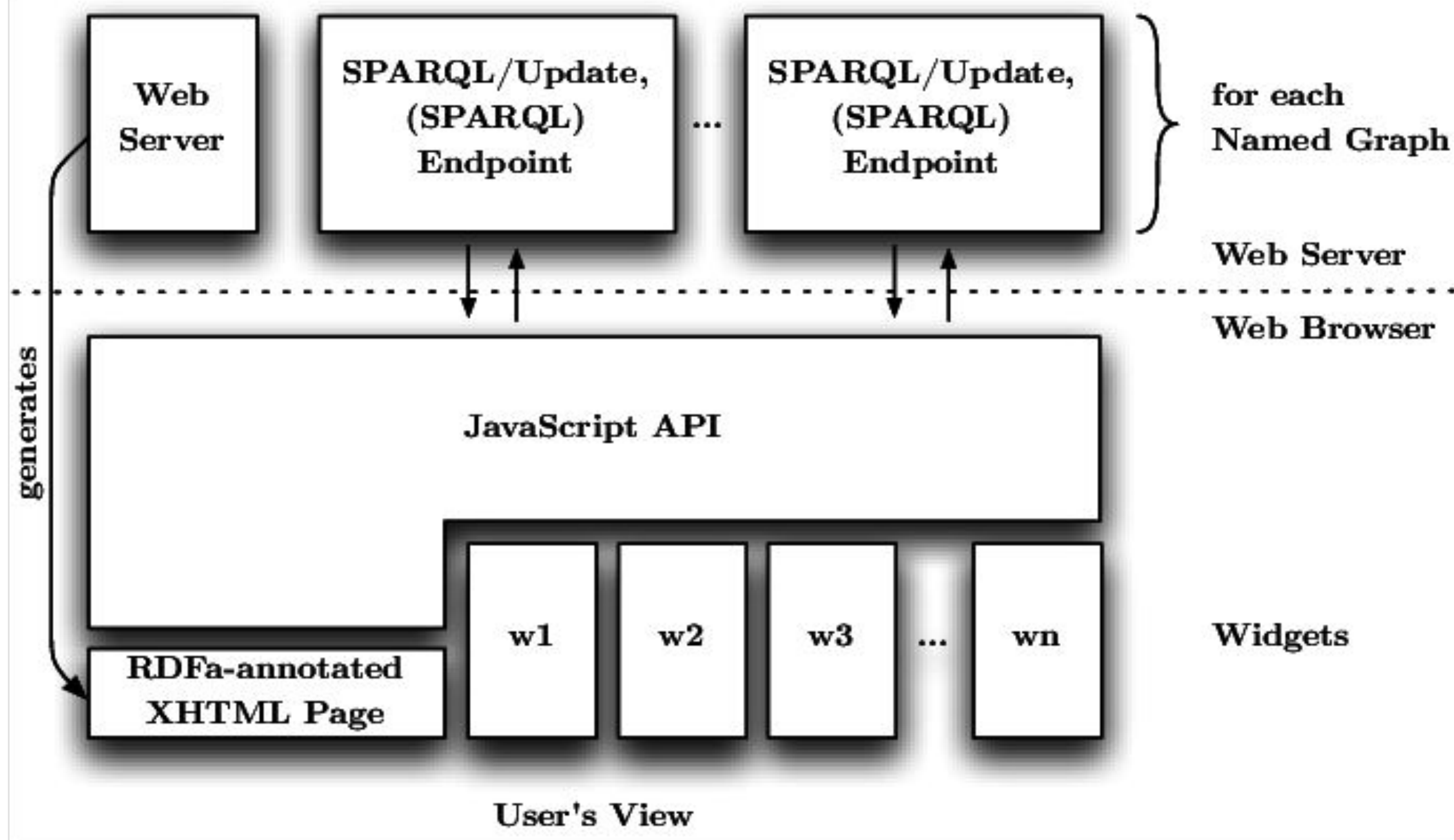




RDFa with HTML microformats

BY

SANA RIZWAN



RDFa

Alice's Blog

www.w3.org/2010/02/rdfa/sources/rdfa-primer/alice-example.html

Delicious LocalData TR 2012 My Mercurial Private Mailing lists Social SW Python RDFa it!

Alice's Blog

W3C RDFa

The trouble with Bob

Date: 10th of September, 2011

Alice

The trouble with Bob is that he takes much better photos than I do:



Beautiful Sunset by Bob.

Alice Birpemsrick,
Email: alice@example.com,
Phone: [+1 617.555.7332](tel:+16175557332).

My buddies:

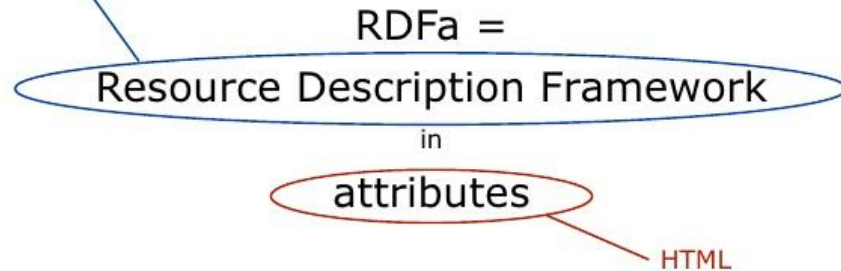
- [Bob](#)
- [Eve](#)
- [Manu](#)

Jo's Barbecue

<https://www.w3.org/TR/rdfa-primer/>

What is RDFa?

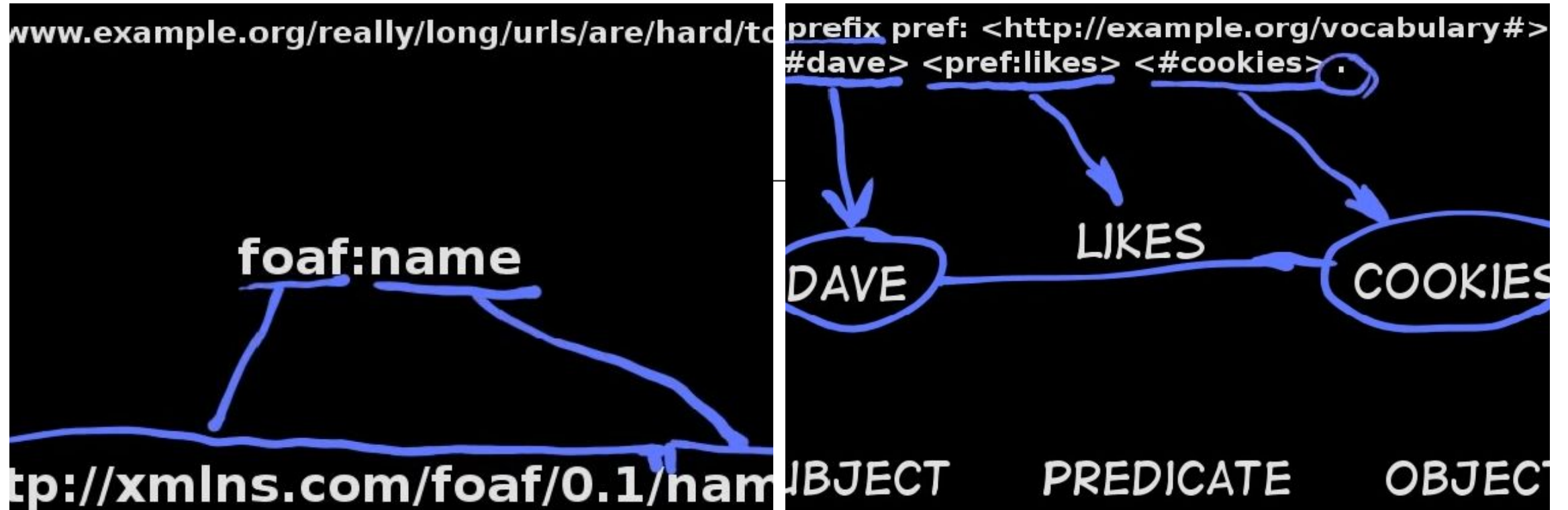
Generic model for the provision of metadata



What is RDFa?

RDFa example

```
<div typeof="foaf:Person" xmlns:foaf="http://xmlns.com/foaf/0.1/">
  <p property="foaf:name">
    Alice Birpemsrick
  </p>
  <p>
    Email: <a rel="foaf:mbox"
href="mailto:alice@example.com">alice@example.com</a>
  </p>
  <p>
    Phone: <a rel="foaf:phone" href="tel:
+1-617-555-7332">+1 617.555.7332</a>
  </p>
</div>
```



FOAF with RDFa

Triple and N3 Notation

```
<body xmlns:foaf="http://xmlns.com/foaf/0.1/">
  <span about="#jane" property="foaf:name">
    Jane McJanerson
  </span>
</body>
```

N3 { @prefix foaf: <http://xmlns.com/foaf/0.1/> .
<#jane> <foaf:name> "Jane McJanerson" .



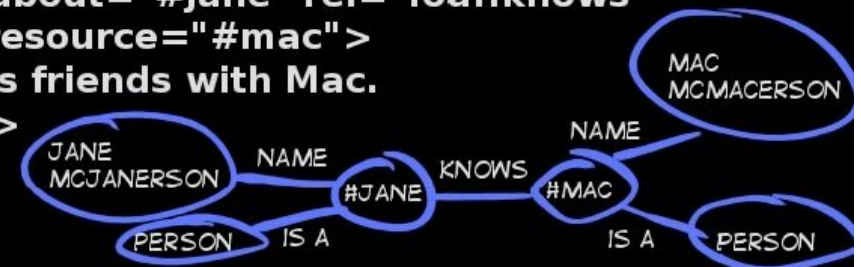
Creating a Triple

```
<body>
  <span>
    </span>
  </body>
```

```
<body xmlns:foaf="http://xmlns.com/foaf/0.1/">
  <span about="#jane" instanceof="foaf:Person"
    property="foaf:name">
    Jane McJanerson
  </span>
```



```
<body xmlns:foaf="http://xmlns.com/foaf/0.1/">
  <span about="#jane" instanceof="foaf:Person"
    property="foaf:name">
    Jane McJanerson
  </span>
  <span about="#mac" instanceof="foaf:Person"
    property="foaf:name">
    Mac McMacerson
  </span>
  <span about="#jane" rel="foaf:knows"
    resource="#mac">
    Jane is friends with Mac.
  </span>
</body>
```





On the left, what browsers see. On the right, what humans see. Can we bridge the gap so that browsers see more of what we see?

RDFa tags/attributes

- **about** attribute determines what we are talking about;
- **property** attribute defines the type of relation between resources;
- **typeof** attribute defines the class to instantiate.
- **about** and **src**: the resource the metadata is about
- **rel** and **rev**: (reverse) relationship between resources
- **href** and **resource**: the partner resource
- **property**: a property for the content of an element
- **content**: override content of an element
- **datatype**: specify the datatype of text
- **typeof**: specifies the RDF type(s) or a subject

XHTML

```
...
All content on this site is licensed under
<a href="http://creativecommons.org/licenses/by/3.0/">
  a Creative Commons License
</a>.
```

XHTML + RDFa

```
...
All content on this site is licensed under
<a rel="license" href="http://creativecommons.org/licenses/by/3.0/">
  a Creative Commons License
</a>.
```

XHTML

```
<div>
  <h2>The trouble with Bob</h2>
  <h3>Alice</h3>
  ...
</div>
```

XHTML + RDFa

```
<div xmlns:dc="http://purl.org/dc/elements/1.1/">
  <h2 property="dc:title">The trouble with Bob</h2>
  <h3 property="dc:creator">Alice</h3>
  ...
</div>
```

XHTML + RDFa

XHTML + RDFa

```
<div xmlns:dc="http://purl.org/dc/elements/1.1/">

  <div about="/alice/posts/trouble_with_bob">
    <h2 property="dc:title">The trouble with Bob</h2>
    <h3 property="dc:creator">Alice</h3>
    ...
  </div>

  <div about="/alice/posts/jos_barbecue">
    <h2 property="dc:title">Jo's Barbecue</h2>
    <h3 property="dc:creator">Eve</h3>
    ...
  </div>

  ...

</div>
```

XHTML + RDFa

```
<div typeof="foaf:Person" xmlns:foaf="http://xmlns.com/foaf/0.1/">
  <p property="foaf:name">
    Alice Birpemswick
  </p>

  <p>
    Email: <a rel="foaf:mbox"
href="mailto:alice@example.com">alice@example.com</a>
  </p>

  <p>
    Phone: <a rel="foaf:phone" href="tel:+1-617-555-7332">+1
617.555.7332</a>
  </p>
</div>
```

Microformats

- A Microformat extends conventional HTML tags with semantic information
- Started by Technorati, Inc.; now community-driven (IRC, mailing list, blogs)
- Make use of the following (X)HTML attributes
 - class
 - rel

Microformats Example

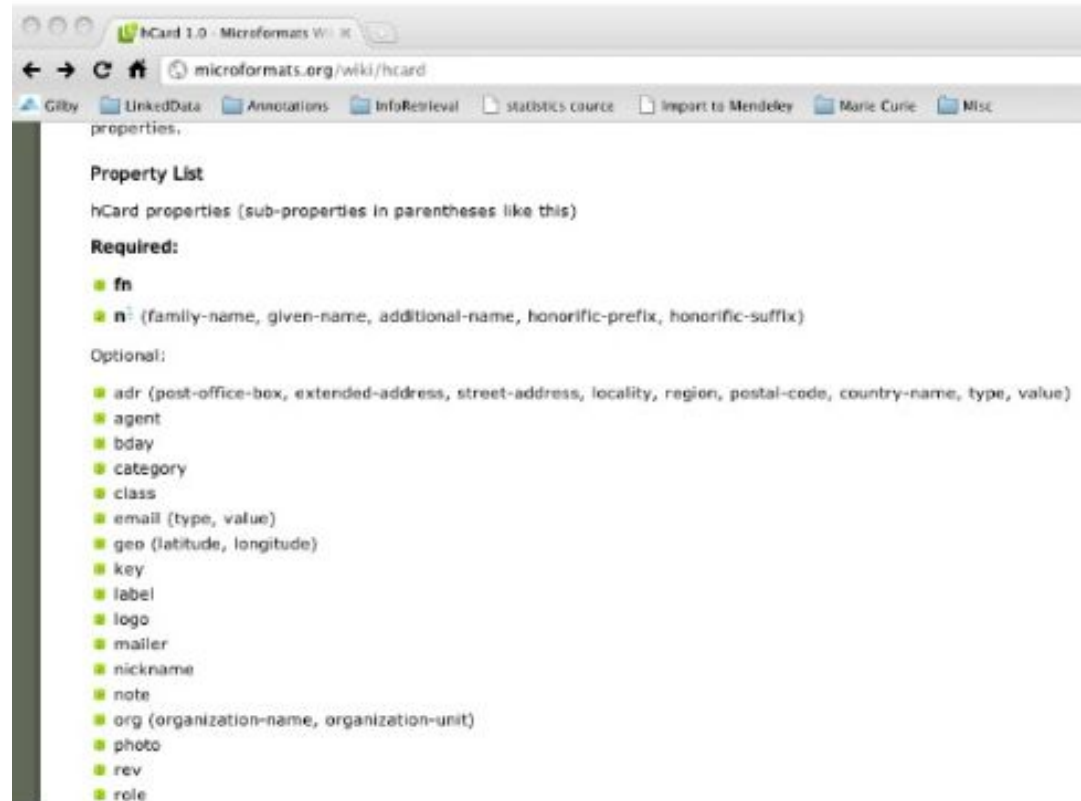
HTML

```
<div>
  <div>Joe Doe</div>
  <div>Jo</div>
  <div>The Example Company</div>
  <div>604-555-1234</div>
  <a href="http://example.com/">http://example.com/</a>
</div>
```

XHTML + Microformats

```
<head profile="http://www.w3.org/2006/03/hcard">
...
</head>
...
<div class="vcard">
  <div class="fn">Joe Doe</div>
  <div class="nickname">Jo</div>
  <div class="org">The Example Company</div>
  <div class="tel">604-555-1234</div>
  <a class="url" href="http://example.com/">http://example.com/</a>
</div>
```


Microformats Example



A screenshot of a web browser showing the 'hCard 1.0 - Microformats Wiki' page. The address bar shows 'microformats.org/wiki/hcard'. The page content includes a 'Property List' section with 'hCard properties (sub-properties in parentheses like this)'. It lists 'Required' properties: 'fn' and 'n' (family-name, given-name, additional-name, honorific-prefix, honorific-suffix). It also lists 'Optional' properties: 'adr' (post-office-box, extended-address, street-address, locality, region, postal-code, country-name, type, value), 'agent', 'bday', 'category', 'class', 'email' (type, value), 'geo' (latitude, longitude), 'key', 'label', 'logo', 'mailer', 'nickname', 'note', 'org' (organization-name, organization-unit), 'photo', 'rev', and 'role'.

hCard 1.0 - Microformats Wiki

microformats.org/wiki/hcard

properties.

Property List

hCard properties (sub-properties in parentheses like this)

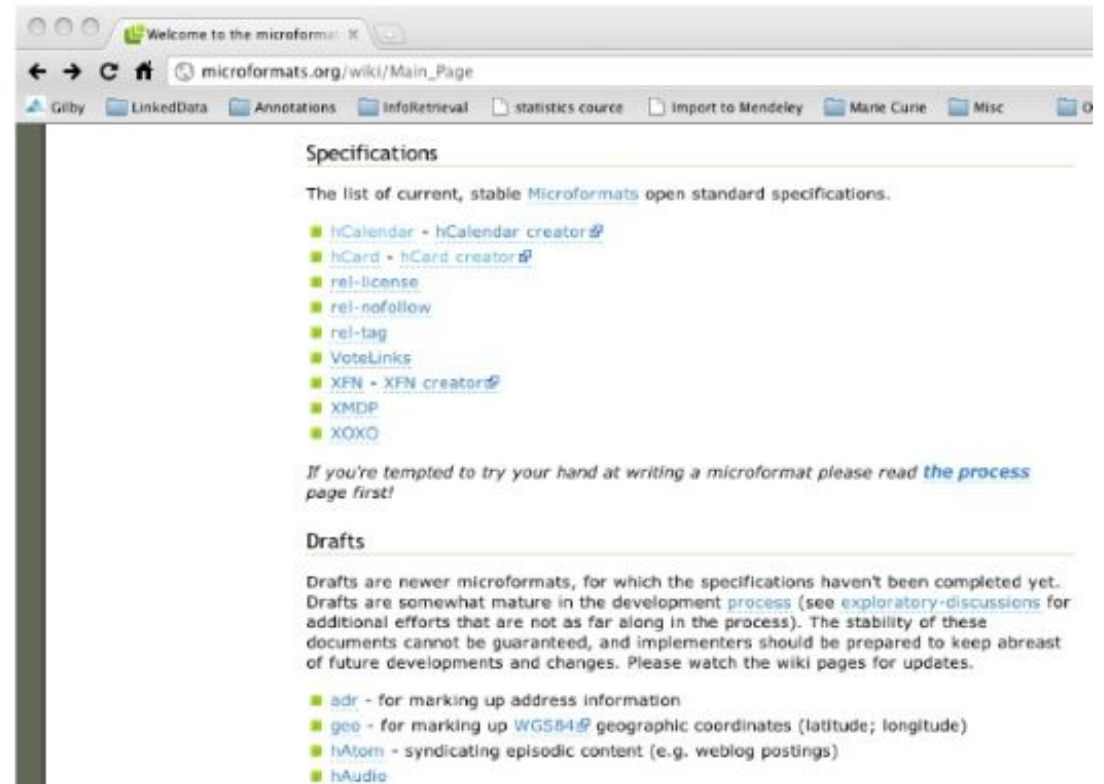
Required:

- fn
- n (family-name, given-name, additional-name, honorific-prefix, honorific-suffix)

Optional:

- adr (post-office-box, extended-address, street-address, locality, region, postal-code, country-name, type, value)
- agent
- bday
- category
- class
- email (type, value)
- geo (latitude, longitude)
- key
- label
- logo
- mailer
- nickname
- note
- org (organization-name, organization-unit)
- photo
- rev
- role

Available Microformats



A screenshot of a web browser showing the 'Welcome to the microformat' page. The address bar shows 'microformats.org/wiki/Main_Page'. The page content includes a 'Specifications' section with a list of current, stable microformats: hCalendar, hCard, rel-license, rel-nofollow, rel-tag, VoteLinks, XFN, XMDP, and XOXO. It also includes a 'Drafts' section with a list of newer microformats: adr, geo, hAtom, and hAudio. A note mentions that drafts are newer microformats for which specifications haven't been completed yet.

Welcome to the microformat

microformats.org/wiki/Main_Page

Specifications

The list of current, stable [Microformats](#) open standard specifications.

- hCalendar - [hCalendar creator](#)
- hCard - [hCard creator](#)
- rel-license
- rel-nofollow
- rel-tag
- VoteLinks
- XFN - [XFN creator](#)
- XMDP
- XOXO

If you're tempted to try your hand at writing a microformat please read [the process page first!](#)

Drafts

Drafts are newer microformats, for which the specifications haven't been completed yet. Drafts are somewhat mature in the development [process](#) (see [exploratory-discussions](#) for additional efforts that are not as far along in the process). The stability of these documents cannot be guaranteed, and implementers should be prepared to keep abreast of future developments and changes. Please watch the wiki pages for updates.

- adr - for marking up address information
- geo - for marking up [WGS84](#) geographic coordinates (latitude; longitude)
- hAtom - syndicating episodic content (e.g. weblog postings)
- hAudio

Microformats vs. RDFa

```
<div xmlns="http://www.w3.org/1999/xhtml"
      xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
      xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
      xmlns:v="http://www.w3.org/2006/vcard/ns#">

  <div about="http://example.com/me/behas" typeof="v:VCard">
    <span property="v:fn">Bernhard Haslhofer</span>
    <span property="v:nickname">behas</span>
    <div rel="v:adr">
      <div typeof="v:Address v:Work">
        <span property="v:street-address">301 College Avenue</span>
        <span property="v:locality">Ithaca</span>,
        <span property="v:postal-code">14850</span>,
        <span property="v:country-name">United States</span>.
      </div>
    </div>
    <a rel="v:email"
href="mailto:bernhard.haslhofer@cornell.edu">bernhard.haslhofer@cornell.edu</a>.
    </div>
  </div>
```

Sample vCard Object in RDFa (see <http://www.w3.org/Submission/vcard-rdf/>)

<https://www.w3.org/Submission/vcard-rdf/>

Microformats vs. RDFa

```
<div class="vcard">

  <span class="fn">Bernhard Haslhofer</span>

  <div class="adr">
    <div class="street-address">301 College Avenue</div>
    <span class="locality">Ithaca</span>
    <span class="postal-code">14850</span>
    <span class="country-name">United States</span>
  </div>

  <a class="email"
    href="mailto:bernhard.haslhofer@cornell.edu">bernhard.haslhofer@cornell.edu</a>

</div>
```

<http://microformats.org/code/hcard/creator>

Card sample created with <http://microformats.org/code/hcard/creator>

Comparison

Microformats	RDFa
flat namespace	XML namespaces
support HTML4, XHTML 1.1, and HTML 5	support for XHTML 1.1
use latent HTML attributes	introduces new metadata attributes
vocabulary defined by one organization/community	open to any RDF-based vocabulary

Adding Machine-Readable Hints to Web Pages

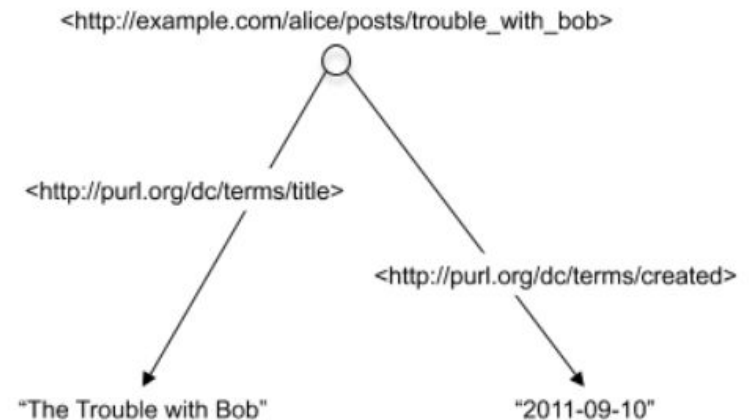
Alice, a blogger who publishes a mix of professional and personal articles at <http://example.com/alice>.

```
<html>
<head>
  ...
</head>
<body>
  ...
  <h2>The Trouble with Bob</h2>
  <p>Date: 2011-09-10</p>
  ...
</body>
```

```
<html>
<head>
  ...
</head>
<body>
  ...
  <h2 property="http://purl.org/dc/terms/title">The Trouble with Bob</h2>
  <p>Date: <span property="http://purl.org/dc/terms/created">2011-09-10</span></p>
  ...
</body>
```

**RDFa
Hints**

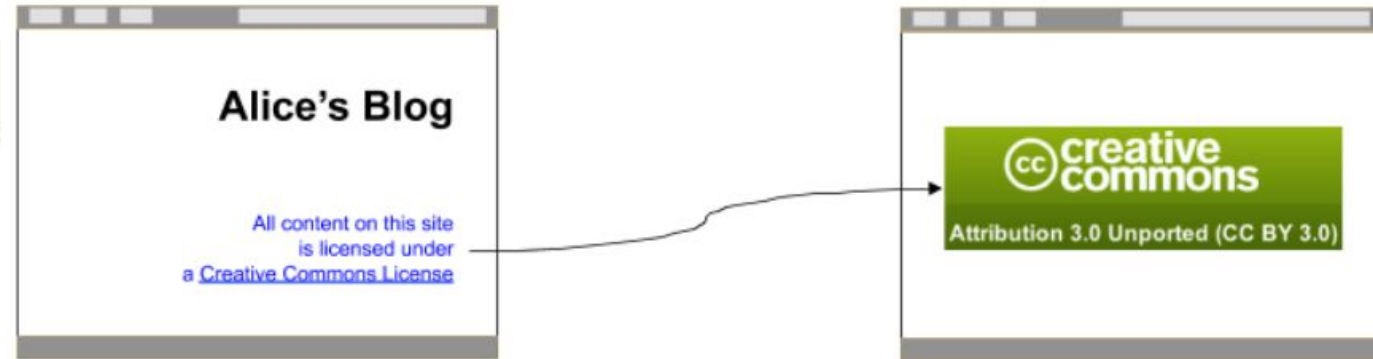
Alice publishes a blog and would like to provide extra structural information on her pages like the publication date or the title. She would like to use the terms defined in the Dublin Core vocabulary.



A visualization of the structured data for a blog post with a title of "The Trouble with Bob" and a creation date.

Links with Flavor

```
<p>All content on this site is licensed under  
<a href="http://creativecommons.org/licenses/by/3.0/">  
  a Creative Commons License</a>. ©2011 Alice Birpemschwick.</p>
```



Add flavor using the **property** attribute. Elements contains the **href/src** attribute



```
<p>All content on this site is licensed under  
<a property="http://creativecommons.org/ns#license" href="http://creativecommons.org/licenses/by/3.0/">  
  a Creative Commons License</a>. ©2011 Alice Birpemschwick.</p>
```


Setting a Default Vocabulary

RDFa introduces the vocab attribute to let the author declare a single vocabulary for a chunk of HTML.

Property values are single "terms" that are concatenated to the URL defined via the vocab attribute. The attribute can be placed on *any* HTML element (i.e., not only on the body element).

```
<html>
<head>
  ...
</head>
<body>
  ...
  <h2 property="http://purl.org/dc/terms/title">The Trouble with Bob</h2>
  <p>Date: <span property="http://purl.org/dc/terms/created">2011-09-10</span></p>
  ...
</body>
```

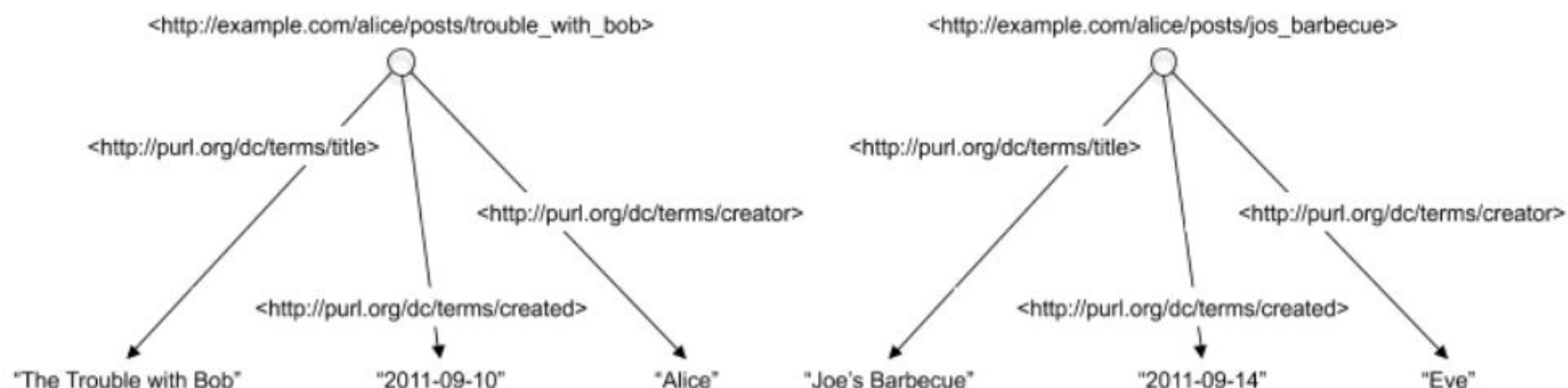
```
<html>
<head>
  ...
</head>
<body vocab="http://purl.org/dc/terms/">
  ...
  <h2 property="title">The Trouble with Bob</h2>
  <p>Date: <span property="created">2011-09-10</span></p>
  ...
</body>
```

```
<html>
<head>
  ...
</head>
<body vocab="http://purl.org/dc/terms/">
  ...
  <h2 property="title">The Trouble with Bob</h2>
  <p>Date: <span property="http://purl.org/dc/terms/created">2011-09-10</span></p>
  ...
</body>
```

```
<html>
<head>
  ...
</head>
<body vocab="http://purl.org/dc/terms/">
  ...
  <h2 property="title">The Trouble with Bob</h2>
  <p>Date: <span property="created">2011-09-10</span></p>
  ...
  <p>All content on this site is licensed under
    <a property="http://creativecommons.org/ns#license" href="http://creativecommons.org/licenses/by/3.0/">
      a Creative Commons License</a>. ©2011 Alice Birpemsrick.</p>
</body>
</html>
```

Multiple Items per Page

RDFa provides **resource**, an attribute for specifying the "context", i.e., the exact URL to which the contained RDFa markup applies.



```
<body vocab="http://purl.org/dc/terms/">
  ...
  <div resource="/alice/posts/trouble_with_bob">
    <h2 property="title">The trouble with Bob</h2>
    <p>Date: <span property="created">2011-09-10</span></p>
    <h3 property="creator">Alice</h3>
    ...
  </div>
  ...
  <div resource="/alice/posts/jos_barbecue">
    <h2 property="title">Jo's Barbecue</h2>
    <p>Date: <span property="created">2011-09-14</span></p>
    <h3 property="creator">Eve</h3>
    ...
  </div>
  ...
</body>
```

Multiple Items per Page: each blog entry is represented by its own node, with properties attached to each.

RDFa concept:

- RDFa shares a number of high-level design principles with Microformats.
- Both use (X)HTML attributes to express the additional data that can be retrieved from the (X)HTML content and converted into RDF, and both rely on external processing to retrieve the necessary information by interpreting the structure of the document.
- RDFa is more flexible: if an RDF vocabulary has been defined by a community for some purpose, that vocabulary can be directly reused by RDFa publishers, whereas Microformats must redefine that vocabulary, through some additional community agreement, into a microformat dialect.

RDFa handling vocabularies:

- RDFa is perfectly well prepared to handle several vocabularies within one file. This is because RDFa is based on the usage of URIs, meaning that usage of various terms can be interleaved without any particular problems.
- Vocabularies defined on the Semantic Web are also based on URIs, this means that RDFa can readily incorporate and benefit of the existence of many vocabularies and ontologies that the Semantic Web community develops, regardless of the complexity of these.

RDFa abbreviate URIs:

- To make the usage of URIs easier, RDFa employs a namespace-like mechanism to abbreviate URIs via strings like `dc:title` or `foaf:Person` to replace full URIs. Users can declare these prefixes using the `xmlns: attributes` on any element in the XHTML hierarchy.
- The “compact URIs,” or **CURIEs**, are used in specific attributes, such as `@rel` and `@property`. When an attribute needs to use either a URI or a CURIE, for example in `@about`, CURIEs are differentiated using square brackets, for example, `[prefix:suffix]`.

Example RDFa + HTML

```
<HTML>
<head><title>An RDFa sample</title></head>
<body>
  <p>The author of <b>UML Distilled</b>
  is <i>Martin Fowler</i></html>
</p>
</body>
</html>
```

Validate it:

<http://rdfa.info/play/>

https://www.w3.org/2012/pyRdfa/Validator.html#distill_by_uri

Metadata in Web-pages and RDFa provides a generic way to do this by building on features already in HTML. Code shows how to enrich the HTML code using RDFa technology

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML+RDFa 1.0//EN"
"http://www.w3.org/Markup/DTD/xhtml-rdfa-1.dtd">
<html xmlns="http://www.w3.org/1999/xhtml"
xmlns:ex="http://example.org/" version="XHTML+RDFa 1.0" xml:lang="en">
  <head><title>An RDFa sample</title></head>
  <body>
    <p><div typeof="ex:Book" about="http://example.org/uml" >
      The author of <b><span property="ex:title">UML Distilled</span></b>
      is <div typeof="ex:Person" about="http://example.org/mfowler">
        <i><span property="ex:name">Martin</span>
        <span property="ex:surname">Fowler</span></div></i>
      </div></p>
    </body>
  </html>
```

THE END

